



3 Complex Analysis 2024

Tutorial 10

A/Prof Elena Berdysheva
Mita Ramabulana



Real integrals

Problem 41 Evaluate the following real integrals using methods of complex analysis:

(a) $\int_0^{2\pi} \frac{1}{1 - 2r \cos \theta + r^2} d\theta$, where $0 \leq r < 1$,

(b) $\int_0^\infty \frac{1}{x^4 + 6x^2 + 8} dx$,

(c) P.V. $\int_{-\infty}^\infty \frac{1}{(2-x)(x^2+4)} dx$,

(d) P.V. $\int_{-\infty}^\infty \frac{2x+3}{x^2+2x+2} dx$.

Solution.

(a) Put

$$I = \int_0^{2\pi} \frac{1}{1 - 2r \cos \theta + r^2} d\theta.$$

Consider the unit circle $\gamma(\theta) = e^{i\theta}$, $\theta \in [0, 2\pi]$. We have $\gamma'(\theta) = ie^{i\theta}$. Using $2 \cos \theta = e^{i\theta} + e^{-i\theta}$, we write

$$\begin{aligned} I &= \int_0^{2\pi} \frac{1}{1 - r(e^{i\theta} + e^{-i\theta}) + r^2} \frac{ie^{i\theta}}{ie^{i\theta}} d\theta = -i \int_\gamma \frac{1}{1 - r\left(z + \frac{1}{z}\right) + r^2} \frac{1}{z} dz \\ &= i \int_\gamma \frac{1}{rz^2 - (r^2 + 1)z + r} dz. \end{aligned}$$

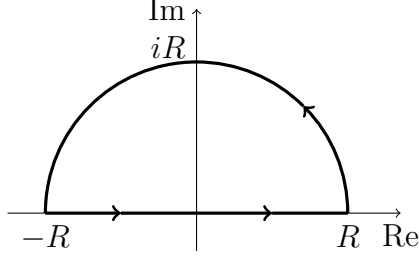
The term $rz^2 - (r^2 + 1)z + r$ has two zeros $z = r$ and $z = \frac{1}{r}$. Only the zero $z = r$ lies inside the unit circle. We have

$$\begin{aligned} I &= 2\pi i \cdot i \operatorname{Res} \left(\frac{1}{rz^2 - (r^2 + 1)z + r}, r \right) = -2\pi \frac{1}{2rz - (r^2 + 1)} \Big|_{z=r} \\ &= -2\pi \frac{1}{2r^2 - r^2 - 1} = \frac{2\pi}{1 - r^2}. \end{aligned}$$

(b) We have

$$\int_0^\infty \frac{1}{x^4 + 6x^2 + 8} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{1}{x^4 + 6x^2 + 8} dx.$$

Consider the path γ indicated in the picture below with R large enough:



Consider the integral

$$\int_{\gamma} \frac{1}{z^4 + 6z^2 + 8} dz.$$

The integrand has four simple poles $\pm 2i, \pm\sqrt{2}i$. Two of them, namely $2i$ and $\sqrt{2}i$ lie in the upper half-plane. Thus,

$$\begin{aligned} \int_{\gamma} \frac{1}{z^4 + 6z^2 + 8} dz &= 2\pi i \operatorname{Res} \left(\frac{1}{z^4 + 6z^2 + 8}, 2i \right) + 2\pi i \operatorname{Res} \left(\frac{1}{z^4 + 6z^2 + 8}, \sqrt{2}i \right) \\ &= 2\pi i \frac{1}{4z^3 + 12z} \Big|_{z=2i} + 2\pi i \frac{1}{4z^3 + 12z} \Big|_{z=\sqrt{2}i} \\ &= 2\pi i \left(-\frac{1}{8i} \right) + 2\pi i \frac{1}{4\sqrt{2}i} = 2\pi \frac{\sqrt{2} - 1}{8}. \end{aligned}$$

If $|z| = R$, then $|z^4 + 6z^2 + 8| \geq |z|^4 - 6|z|^2 - 8 \geq \frac{1}{2}R^4$, if R is large enough. We estimate the integral over the semicircle γ_R as follows:

$$\left| \int_{\gamma_R} \frac{1}{z^4 + 6z^2 + 8} dz \right| \leq \frac{2}{R^4} \cdot \pi R = \frac{2\pi}{R^3} \rightarrow 0, \quad R \rightarrow \infty.$$

Since

$$\int_{\gamma} \frac{1}{z^4 + 6z^2 + 8} dz = \left(\int_{[-R, R]} + \int_{\gamma_R} \right) \frac{1}{z^4 + 6z^2 + 8} dz,$$

we obtain

$$\int_{-\infty}^{\infty} \frac{1}{x^4 + 6x^2 + 8} dx = \lim_{R \rightarrow \infty} \int_{[-R, R]} \frac{1}{z^4 + 6z^2 + 8} dz = 2\pi \frac{\sqrt{2} - 1}{8}.$$

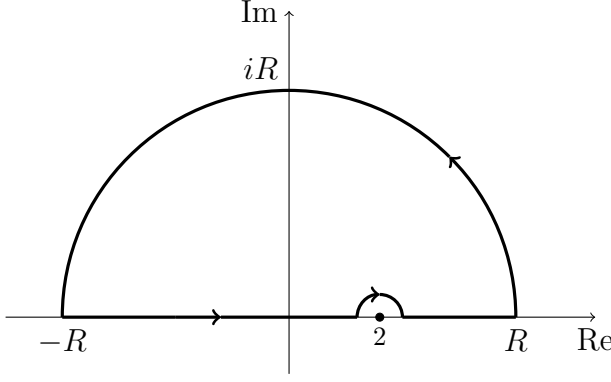
and

$$\int_0^{\infty} \frac{1}{x^4 + 6x^2 + 8} dx = \pi \frac{\sqrt{2} - 1}{8}.$$

(c) The integral converges at ∞ , but has a pole $x = 2$ on the real line. We have

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1}{(2-x)(x^2+4)} dx = \lim_{\substack{R \rightarrow \infty \\ \varepsilon \rightarrow 0+}} \left(\int_{[-R, 2-\varepsilon]} + \int_{[2+\varepsilon, R]} \right) \frac{1}{(2-z)(z^2+4)} dz.$$

Consider the path γ indicated below with R large enough:



Consider

$$\int_{\gamma} \frac{1}{(2-z)(z^2+4)} dz.$$

The integrand has three simple poles $z = 2$, $z = \pm 2i$. Only the pole $z = 2i$ lie inside γ . We have

$$\begin{aligned} \int_{\gamma} \frac{1}{(2-z)(z^2+4)} dz &= 2\pi i \operatorname{Res} \left(\frac{1}{(2-z)(z+2i)(z-2i)}, 2i \right) \\ &= 2\pi i \frac{1}{(2-2i)(2i+2i)} = \frac{\pi(1+i)}{8}. \end{aligned}$$

If $|z| = R$, then for R large enough

$$|(2-z)(z^2+4)| = |-z^3 + 2z^2 - 4z + 8| \geq |z|^3 - 2|z|^2 - 4|z| - 8 \geq \frac{1}{2}R^3.$$

We estimate the integral over the large semicircle γ_R as follows:

$$\left| \int_{\gamma_R} \frac{1}{(2-z)(z^2+4)} dz \right| \leq \frac{2}{R^3} \pi R \rightarrow 0, \quad R \rightarrow \infty.$$

For the integral over the small semicircle γ_{ε} centered at 2 (negatively oriented) we have

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0+} \int_{\gamma_{\varepsilon}} \frac{1}{(2-z)(z^2+4)} dz &= -\frac{1}{2} 2\pi i \operatorname{Res} \left(\frac{1}{(2-z)(z^2+4)}, 2 \right) = -\pi i \frac{1}{2^2+4} \\ &= -\pi i \lim_{z \rightarrow 2} (z-2) \frac{1}{(2-z)(z^2+4)} = \frac{\pi i}{8}. \end{aligned}$$

Since

$$\int_{\gamma} \frac{1}{(2-z)(z^2+4)} dz = \left(\int_{[-R, 2-\varepsilon]} + \int_{\gamma_{\varepsilon}} + \int_{[2+\varepsilon, R]} + \int_{\gamma_R} \right) \frac{1}{(2-z)(z^2+4)} dz,$$

we obtain

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1}{(2-x)(x^2+4)} dx = \frac{\pi(1+i)}{8} - \frac{\pi i}{8} = \frac{\pi}{8}.$$

- (d) The integrand has no poles on the real axis, but the integral does not converge at ∞ . We use the same path as in (b) and consider the integral

$$\int_{\gamma} \frac{2z+3}{z^2+2z+2} dz.$$

The integrand has two simple poles $z = -1 \pm i$, only the pole $z = -1+i$ lies inside γ . Thus,

$$\int_{\gamma} \frac{2z+3}{z^2+2z+2} dz = 2\pi i \operatorname{Res} \left(\frac{2z+3}{z^2+2z+2}, -1+i \right) = 2\pi i \left. \frac{2z+3}{2z+2} \right|_{z=-1+i} = \pi(1+2i).$$

Notice that the integral over the semicircle $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$, does not tend to zero as $R \rightarrow \infty$. However, we can calculate its limit:

$$\begin{aligned} \int_{\gamma_R} \frac{2z+3}{z^2+2z+2} dz &= \int_0^{\pi} \frac{2Re^{it}+3}{R^2e^{i2t}+2Re^{it}+2} iRe^{it} dt \\ &= i \int_0^{\pi} \frac{2R^2e^{i2t}+3Re^{it}}{R^2e^{i2t}+2Re^{it}+2} dt \\ &= i \int_0^{\pi} \frac{2+\frac{3}{R}e^{-it}}{1+\frac{2}{R}e^{-it}+\frac{2}{R^2}e^{-i2t}} dt \\ &\rightarrow i \int_0^{\pi} 2 dt = 2\pi i. \end{aligned}$$

The interchange of the limit and the integral above is justified by the fact that $\frac{2+\frac{3}{R}e^{-it}}{1+\frac{2}{R}e^{-it}+\frac{2}{R^2}e^{-i2t}}$ converges to 2 uniformly on $[0, \pi]$ (see Problem 24 (a)).

We conclude that

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{2x+3}{x^2+2x+2} dx = \pi(1+2i) - 2\pi i = \pi.$$

Laurent series

Problem 42 Find the Laurent expansion of the function $f(z) = \frac{1}{z^2-3z+2}$ in each of the following regions:

- (a) $\{z \in \mathbb{C} : |z| < 1\}$,
- (b) $\{z \in \mathbb{C} : 1 < |z| < 2\}$,
- (c) $\{z \in \mathbb{C} : |z| > 2\}$,
- (d) $\{z \in \mathbb{C} : 0 < |z-1| < 1\}$,
- (e) $\{z \in \mathbb{C} : |z-1| > 1\}$.

Solution.

We have

$$f(z) = \frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)} = \frac{1}{1-z} - \frac{1}{2-z}.$$

(a) If $|z| < 1$, then

$$f(z) = \frac{1}{1-z} - \frac{1}{2-z} = \frac{1}{1-z} - \frac{1}{2} \frac{1}{1-\frac{z}{2}} = \sum_{n=0}^{\infty} z^n - \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n.$$

(b) If $1 < |z| < 2$, then

$$f(z) = -\frac{1}{z} \frac{1}{1-\frac{1}{z}} - \frac{1}{2} \frac{1}{1-\frac{z}{2}} = -z^{-1} \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n - \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n = -\sum_{n=1}^{\infty} z^{-n} - \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} z^n.$$

(c) If $|z| > 2$, then

$$\begin{aligned} f(z) &= -\frac{1}{z} \frac{1}{1-\frac{1}{z}} + \frac{1}{z} \frac{1}{1-\frac{2}{z}} = -\sum_{n=1}^{\infty} z^{-n} + z^{-1} \sum_{n=0}^{\infty} \left(\frac{2}{z}\right)^n = -\sum_{n=1}^{\infty} z^{-n} + \sum_{n=0}^{\infty} 2^n z^{-n-1} \\ &= -\sum_{n=1}^{\infty} z^{-n} + \sum_{n=1}^{\infty} 2^{n-1} z^{-n} = \sum_{n=2}^{\infty} (2^{n-1} - 1) z^{-n}. \end{aligned}$$

(d) If $0 < |z-1| < 1$, then

$$f(z) = -\frac{1}{z-1} - \frac{1}{1-(z-1)} = -(z-1)^{-1} - \sum_{n=0}^{\infty} (z-1)^n.$$

(e) If $|z-1| > 1$, then

$$\begin{aligned} f(z) &= -\frac{1}{z-1} + \frac{1}{(z-1)-1} = -(z-1)^{-1} + \frac{1}{z-1} \frac{1}{1-\frac{1}{z-1}} \\ &= -(z-1)^{-1} + (z-1)^{-1} \sum_{n=0}^{\infty} \left(\frac{1}{z-1}\right)^n = -(z-1)^{-1} + \sum_{n=0}^{\infty} (z-1)^{-n-1} \\ &= -(z-1)^{-1} + \sum_{n=1}^{\infty} (z-1)^{-n} = \sum_{n=2}^{\infty} (z-1)^{-n-1}. \end{aligned}$$

Problem 43 Expand each of the functions in a Laurent series about the indicated point z_0 . Use the Laurent series to calculate $\text{Res}(f, z_0)$.

(a) $f(z) = \frac{\sin z}{z^4}$, $z_0 = 0$,

(b) $f(z) = \frac{e^z}{z^2 - 1}$, $z_0 = 1$,

(c) $f(z) = \sin\left(\frac{1}{z}\right)$, $z_0 = 0$.

Solution.

(a) We have

$$f(z) = \frac{\sin z}{z^4} = z^{-4} \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n-3}}{(2n+1)!}$$

The series converges for $0 < |z| < \infty$. The coefficient of the power z^{-1} is the coefficient with $n = 1$, and

$$\text{Res}(f, 0) = (-1) \cdot \frac{1}{3!} = -\frac{1}{6}.$$

(b) We are trying to find the Laurent expansion in the annulus $0 < |z - 1| < 2$. We have

$$f(z) = \frac{e^z}{z^2 - 1} = \frac{e}{z - 1} e^{z-1} \frac{1}{z + 1}.$$

The power series expansion of e^{z-1} is

$$e^{z-1} = \sum_{n=0}^{\infty} \frac{1}{n!} (z - 1)^n = 1 + (z - 1) + \frac{1}{2}(z - 1)^2 + \dots.$$

The power series expansion of $\frac{1}{z+1}$ is

$$\begin{aligned} \frac{1}{z+1} &= \frac{1}{(z-1)+2} = \frac{1}{2} \frac{1}{1 - \left(-\frac{z-1}{2}\right)} = \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{z-1}{2}\right)^n \\ &= \frac{1}{2} \left(1 - \frac{1}{2}(z-1) + \frac{1}{4}(z-1)^2 + \dots\right). \end{aligned}$$

Altogether we have

$$\begin{aligned} f(z) &= \frac{e}{2} (z-1)^{-1} \left(1 + (z-1) + \frac{1}{2}(z-1)^2 + \dots\right) \left(1 - \frac{1}{2}(z-1) + \frac{1}{4}(z-1)^2 + \dots\right) \\ &= \frac{e}{2} (z-1)^{-1} + \frac{e}{4} + \frac{e}{8}(z-1) + \frac{e}{48}(z-1)^2 + \dots. \end{aligned}$$

In particular,

$$\text{Res}(f, 0) = \frac{e}{2}.$$

(c) Notice that this function has an essential singularity at 0. We have

$$f(z) = \sin\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left(\frac{1}{z}\right)^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{-2n-1}.$$

In particular,

$$\text{Res}(f, 0) = 1.$$